A P P L I E D A I C L A S S

# Student Reference Guide

Tools · Concepts · Career Applications

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# Part 1: What Is AI? A Quick Primer

This section covers the fundamental concepts you need to understand before diving into tools. You don't need to memorize this — just read it once so the tools make sense when you use them.

## 1.1 Artificial Intelligence: The 60-Second Version

Artificial Intelligence is software that learns patterns from data to make decisions, generate content, or solve problems — without being explicitly programmed for every case.

There are several levels of AI:

- Machine Learning: Systems that learn from examples (spam filters, recommendation engines)
- Deep Learning: Neural networks with many layers (image recognition, voice assistants)
- Large Language Models (LLMs): Trained on billions of words to understand and generate text (ChatGPT, Claude, Gemini)
- AI Agents: LLMs equipped with tools, memory, and the ability to take actions (Claude Cowork, Copilot, AutoGPT)

## 1.2 Tokenization: How AI Reads Your Words

When you type a message to an AI, it doesn't read your words the way a human does. It first converts everything into tokens — small pieces of text, roughly 3/4 of a word each.

**Example: Tokenizing a Simple Sentence**

Input: "Hello, how are you today?"

Tokens: ["Hello"] [","] ["how"] ["are"] ["you"] ["today"] ["?"]

Token IDs: [15496] [11] [4919] [389] [345] [1909] [30]

Embeddings: Each ID becomes a 768-dimensional vector of numbers encoding meaning.

Key facts:

- GPT-4 can process up to 128,000 tokens at once (~96,000 words)
- Claude handles up to 200,000 tokens
- Gemini Ultra processes up to 2 million tokens
- One token ≈ ¾ of a word in English

Those token IDs then become vectors (embeddings) — lists of numbers that represent the meaning of each word in mathematical space. Words with similar meanings land close together. 'King' and 'Queen' are close; 'Dog' and 'Democracy' are far apart.

Why this matters for you: the context window (how many tokens the AI can see at once) determines how much content you can analyze in one session. If you're analyzing a 100-page document, you need a model with a large context window.

## 1.3 Neural Networks: The Engine

An LLM is built on a deep neural network — specifically the Transformer architecture, first described in a 2017 Google paper titled 'Attention Is All You Need.'

Here's the simplified version:

1. Your text is tokenized and converted to vectors
2. Those vectors flow through 80-120 stacked Transformer blocks
3. Each block runs self-attention (every token decides what to 'attend to' in other tokens) plus a feed-forward network
4. The final layer outputs probabilities over all possible next tokens
5. The highest-probability token is selected — and this repeats, one token at a time, until the response is complete

**Scale Context**

GPT-4: ~1.8 trillion parameters (weights) | 120 Transformer layers | 128k context

Claude 3.5 Sonnet: ~200B parameters | 200k context

Gemini Ultra: ~1 trillion parameters | 2M context

A 'parameter' is a learned weight in the network. More parameters = more knowledge capacity.

Bigger is not always better — smaller, well-tuned models often outperform larger ones on specific tasks.

## 1.4 What Does an LLM Actually Do?

The core function of an LLM is deceptively simple: predict the most likely next token, one at a time. At massive scale, with vast training data, this produces systems that can write, reason, code, analyze, and create.

| What LLMs CAN Do | Current Limitations to Know |
|---|---|
| Generate fluent, coherent long-form text | Training data has a cutoff date (may be outdated) |
| Summarize documents instantly | Can hallucinate confident-sounding false information |
| Translate between 100+ languages | No real-time internet by default (unless tool-equipped) |

| | |
|---|---|
| Write code in any programming language | Struggles with precise numerical calculations |
| Answer complex questions from context | No persistent memory by default across sessions |
| Extract structured data from unstructured text | Sensitive to how questions are worded |
| Reason step-by-step (chain-of-thought) | Cannot verify its own outputs |
| Analyze images, audio, and video (multimodal models) | Context window limits how much it can process |

# Part 2: Prompting — The Most Important Skill

Prompting is the art and science of communicating with AI models. It is the single most impactful skill you can develop for getting value from these tools. The gap between a weak prompt and a strong prompt can be the difference between a generic, useless response and a precisely tailored, immediately useful one.

## 2.1 The CRAFT Framework

Every great prompt contains five elements. Use this as a checklist before submitting any important prompt:

| | | | |
|---|---|---|---|
| **C** | **Context** | Who you are, what situation, what background the AI needs | *"I am a Finance VP preparing a Q1 board report for a $50M manufacturing firm..."* |
| **R** | **Role** | What expert persona should the AI adopt? | *"You are a senior financial analyst at a Fortune 500 firm..."* |
| **A** | **Action** | Specific verb: Write / Analyze / Compare / Build / Debug | *"Write a 3-paragraph executive summary highlighting revenue, cost drivers, and 2 key risks..."* |
| **F** | **Format** | Table / Bullets / Email / JSON / 200 words max / no bullet points | *"Return results in a 3-column table: Risk \| Likelihood \| Mitigation..."* |
| **T** | **Tone** | Formal / Casual / Executive / Technical / Persuasive / Friendly | *"Use formal, concise language appropriate for a board of directors."* |

## 2.2 Advanced Prompting Techniques

- Chain-of-thought: Add 'Think step by step' to improve reasoning accuracy on complex problems
- Self-critique: After getting an answer, ask 'What are the 3 weakest points of your previous response? How would you fix them?'
- Few-shot examples: Show the format you want with 1-3 examples before your actual request
- Negative constraints: Tell the AI what NOT to do ('Do NOT use the word leverage. Do NOT start with Certainly!')
- Persona stacking: Combine two expert lenses ('You are both a healthcare attorney AND a product manager...')

## 2.3 Common Mistakes to Avoid

**Six Mistakes That Undermine AI Output**

1. Treating AI output as final — always review, fact-check, and edit. AI is a first-draft machine.
2. Using vague 1-line prompts — add context, role, format, and tone every time.
3. Pasting confidential client data — anonymize or use your company's enterprise AI tool.
4. Expecting AI to always be right — LLMs hallucinate. Verify facts and citations.
5. Using one tool for everything — different tools have different strengths.
6. Ignoring memory/Projects features — set up Custom Instructions and Projects to save time.

# Part 3: Tool-by-Tool Walkthroughs

## 3.1 ChatGPT — Your All-Purpose AI Assistant

**URL:** chatgpt.com | Cost: Free tier + Plus ($20/mo) + Team/Enterprise

ChatGPT is the most widely used AI assistant in the world. GPT-4o handles text, images, files, and code. The o1 model excels at complex reasoning tasks.

### How to Use ChatGPT — Step by Step

6. Go to chatgpt.com and sign in (or create a free account)
7. Select your model: GPT-4o (balanced, most versatile), o1 (deep reasoning), GPT-4o mini (fast, cheap)
8. Type your prompt or drag-and-drop a file (PDF, image, Excel, etc.)
9. Enable Web Search (toggle in toolbar) for current information
10. Enable Code Interpreter (also in toolbar) to analyze data and create charts from your files
11. Use Projects (left sidebar) to create persistent assistants with custom instructions and uploaded reference documents
12. Set Custom Instructions (Settings) to tell ChatGPT your role, industry, and preferences — applies to every conversation

**Power User Tips**

Create a custom GPT for your specific workflow: top-right menu → Explore GPTs → Create

In Projects, upload your company style guide, templates, or reference docs once — they stay in context forever

For data analysis: upload an Excel file and ask 'Show me the trend and create a chart'

For long documents: paste the full text and ask specific questions rather than asking for a general summary

## 3.2 Claude — Best for Long Documents and Nuanced Reasoning

**URL:** claude.ai | Cost: Free tier + Pro ($20/mo) + Enterprise

Claude is Anthropic's AI assistant. It consistently scores at the top of coding, reasoning, and long-context benchmarks. Its 200,000-token context window means you can paste in entire books, contracts, or datasets.

### How to Use Claude — Step by Step

13. Go to claude.ai and create a free account

14. Select your model: Claude Sonnet 4.6 (best balance), Opus 4.6 (hardest tasks), Haiku (fast, lightweight)
15. Create a Project (left sidebar) — upload reference documents that stay in every conversation in that project
16. Paste or upload large documents — Claude can handle entire contracts, reports, or codebases
17. Use extended thinking mode for complex analysis: ask it to 'think carefully before responding'
18. Use Claude Cowork (the desktop app, which you're using now) to create files, slides, and documents

**Power User Tips**

Projects are Claude's superpower — create one project per client, topic, or workflow

For legal/financial docs: paste the full document and ask 'What are the 5 riskiest clauses for me as the buyer?'

For coding: Claude can write, review, debug, and explain code in any language — ask it to explain what it's doing

Extended thinking: add 'Think this through step by step, consider edge cases, then give me your answer' to complex prompts

## 3.3 Microsoft Copilot — AI Embedded in Your Microsoft 365

**URL:** copilot.microsoft.com | Cost: Free + M365 Copilot ($30/user/mo)

Copilot is embedded directly into Word, Excel, PowerPoint, Teams, and Outlook. It's the best choice if you're already in the Microsoft 365 ecosystem.

### How to Use Copilot in Each App

- In Word: Click the Copilot button → 'Draft a memo about...' or select text → Rewrite / Summarize / Improve tone
- In Excel: Select data → Click Copilot → 'Show trends', 'Add a column that calculates...', 'Highlight anomalies'
- In PowerPoint: Home → Copilot → 'Create a presentation about [topic]' or 'Turn this Word doc into slides'
- In Teams: After any meeting, open the Copilot pane → Ask for 'Summary', 'Action items', or 'What did [person] say about [topic]?'
- In Outlook: Open any email thread → Click Copilot → 'Draft a reply that...', 'Summarize this thread', 'Change the tone to...'

**Power User Tips**

In Excel: use natural language to write complex formulas — 'Add a column that shows 3-month rolling average'

Copilot Pages: Share AI-generated content as live, collaborative documents (like a smarter Google Doc)

Teams Copilot: If you missed a meeting, ask Copilot to brief you on what happened and what actions you need to take

For sensitive data: Copilot keeps your data within the Microsoft 365 environment — safer than public AI tools

## 3.4 Google Gemini — Deep Research and Google Workspace

**URL:** gemini.google.com | Cost: Free + Advanced ($20/mo)

Gemini is Google's flagship AI. The Deep Research feature is particularly powerful — it autonomously searches, synthesizes, and writes comprehensive reports. The 2M token context window is the largest available.

### How to Use Gemini — Step by Step

19. Go to gemini.google.com and sign in with your Google account
20. Try Deep Research: click the 'Deep Research' option and describe your research topic — it will research and write a full report
21. Create Gems: Create → New Gem → set instructions for a specialized assistant (e.g., 'You are my strategy research assistant')
22. Upload images, PDFs, or videos for multimodal analysis
23. Connect to Google Workspace: ask it to search your Gmail, summarize Drive documents, or add Calendar events
24. Use inside Google Docs/Sheets: click the Gemini icon to draft, summarize, or generate formulas

**Power User Tips**

Deep Research is genuinely impressive — use it for competitive intelligence, market sizing, regulatory research

The 2M token context can process your entire company's financial history, a full academic paper corpus, or a massive codebase

Gemini in Google Sheets: highlight a column of unstructured text and ask 'Categorize each entry as Positive, Neutral, or Negative'

Gems vs. ChatGPT Custom GPTs: similar concept, both free with basic tier

## 3.5 Perplexity — AI Research with Citations

**URL:** perplexity.ai | Cost: Free + Pro ($20/mo)

Perplexity is an AI-powered search engine that gives you answers with numbered citations pointing to real sources. It's the tool to use for research that requires current, verifiable information.

### How to Use Perplexity — Step by Step

25. Go to perplexity.ai and ask your research question in natural language
26. Change Focus mode: Academic (peer-reviewed papers), YouTube (video transcripts), Reddit (community discussions), LinkedIn
27. Click any numbered citation in the answer to read the full source document
28. Use Pro Search (click the Pro toggle) for multi-step research with clarifying questions before generating the report
29. Upload a PDF or image (Pro) to ask questions about your specific document
30. Create Spaces to organize and share research with your team

**Power User Tips**

Rule of thumb: Use Perplexity for current events, facts, and research. Use Claude or ChatGPT for analysis and writing.

Academic mode is excellent for literature reviews — it only pulls from peer-reviewed sources

Always copy the citations into your own documents to maintain credibility in deliverables

Competitive intelligence: 'What are the top 5 competitors to [company] and what are their pricing models?' — with sources

## 3.6 Midjourney — World-Class AI Image Generation

**URL:** midjourney.com | Cost: Basic $10/mo, Standard $30/mo

Midjourney produces the highest-quality AI-generated images available. It's used by marketing teams, designers, and presenters to create stunning visuals without a graphic designer.

### How to Use Midjourney — Step by Step

31. Sign up at midjourney.com (Discord is optional now — use the website)
32. Click Create and type a detailed description of your image
33. Add parameters after your description: --ar 16:9 (widescreen), --style raw (photorealistic), --v 6 (latest model)
34. Generate 4 variations — click U1/U2/U3/U4 to upscale your favorite to HD
35. Click V1-V4 to create variations of a specific image
36. Right-click → Save image → insert into PowerPoint, Word, or anywhere

**Prompt Formula That Works**

Structure: "[Subject], [Setting/Background], [Lighting], [Style], [Mood], --ar [ratio] --v 6"

Example: "Professional portrait of a confident executive, modern glass office at sunset, dramatic side lighting, photorealistic, corporate, warm tones --ar 16:9 --style raw --v 6"

Example: "Abstract data visualization of global financial flows, neon blue and purple on dark background, futuristic, ultra-detailed --ar 16:9 --v 6"

Use ChatGPT to help write better Midjourney prompts: 'Write me 3 Midjourney prompts for a hero image about AI in finance'

## 3.7 Replit — Build Apps Without Coding

**URL:** replit.com | Cost: Free tier + Core $15/mo

Replit Agent lets you build functional web applications, dashboards, and tools using plain English. You describe what you want; the AI writes the code, runs it, debugs errors, and gives you a live URL. No coding required.

### How to Use Replit — Step by Step

37. Go to replit.com and create an account
38. Click 'Replit Agent' (NOT 'Create Repl' — that's the manual coding path)
39. Describe your app in detail: 'Build a web app that tracks weekly expenses by category and displays a pie chart. Include a form to add new expenses with date, amount, and category fields.'
40. Watch the Agent write code, install dependencies, and run your app in real time
41. Test your app in the preview panel that appears alongside the editor
42. If something's wrong, just describe the problem: 'The chart isn't showing colors' or 'The form doesn't submit'
43. Click Deploy when satisfied — you get a live URL to share immediately

**Ideas for What to Build**

Expense tracker with category breakdown charts

Project status dashboard with traffic-light indicators

Client meeting notes organizer with search

KPI calculator with scenario modeling (change assumptions, see output)

Team capacity planner (who's available, what's their current load)

Simple CRM for tracking client interactions and follow-ups

Portfolio return calculator with historical comparison

## 3.8 Google AI Studio & NotebookLM — Two Hidden Gems

**AI Studio:** aistudio.google.com | **NotebookLM:** notebooklm.google.com | Both FREE

Google AI Studio gives you free access to the most powerful Gemini models with a 1M+ token context window — ideal for analyzing massive files or building with the Gemini API. NotebookLM creates an AI assistant that reads only your specific documents, eliminating hallucination entirely.

### NotebookLM: Your Personal Research Assistant

44. Go to notebooklm.google.com and sign in with Google
45. Create a new Notebook and give it a descriptive name
46. Add sources: upload PDFs, link Google Docs, paste URLs, add YouTube video links, or upload audio files
47. Ask questions — every answer cites the exact source and page number
48. Generate an Audio Overview: click the headphones icon for a podcast-style briefing on your documents
49. Share your notebook with colleagues for collaborative research

**NotebookLM Killer Use Cases**

Board prep: Upload all board materials → ask 'What are the top 3 risks I should raise tomorrow?'

Contract review: Upload 5 vendor contracts → ask 'Which contract has the most restrictive liability clause?'

Competitive research: Upload competitor annual reports → ask 'What are their stated growth priorities?'

Meeting prep: Upload the last 6 meeting decks → ask 'What commitments did we make in Q3 that we haven't addressed?'

Regulatory compliance: Upload your policy documents and the regulation → ask 'Where are we non-compliant?'

# Part 4: Master Resource List

All tools, links, and resources in one place. Bookmark this page.

| Tool | Link | Cost | Best For |
|---|---|---|---|
| **ChatGPT** | chatgpt.com | Free / $20+/mo | All-purpose writing, analysis, coding, image generation |
| **Claude** | claude.ai | Free / $20+/mo | Long docs, nuanced reasoning, coding, agents |
| **Microsoft Copilot** | copilot.microsoft.com | Free / $30/mo M365 | Word, Excel, PowerPoint, Teams, Outlook |
| **Google Gemini** | gemini.google.com | Free / $20/mo | Deep research, multimodal, Google Workspace |
| **Perplexity** | perplexity.ai | Free / $20/mo | Research with citations, current events, fact-checking |
| **NotebookLM** | notebooklm.google.com | Free | Q&A over your own documents (no hallucination) |
| **Midjourney** | midjourney.com | $10-$30/mo | Professional image generation for marketing & decks |
| **Replit** | replit.com | Free / $15/mo | Build apps, tools, and dashboards without coding |
| **Google AI Studio** | aistudio.google.com | Free | Gemini API, huge context, developer playground |
| **ElevenLabs** | elevenlabs.io | Free / $5+/mo | AI voice cloning and text-to-speech |
| **Runway ML** | runwayml.com | Free / $12+/mo | AI video generation and editing |
| **Cursor** | cursor.com | Free / $20/mo | AI-powered code editor (VS Code with AI built in) |
| **v0 by Vercel** | v0.dev | Free / $20/mo | Generate UI components and web pages from descriptions |
| **Suno** | suno.com | Free / $8+/mo | AI music generation from text descriptions |

# Part 5: Your 30-Day AI Action Plan

The professionals who get the most from AI are not the most technical — they are the most consistent. Here is a concrete 30-day plan to go from beginner to confident AI user.

## This Week: Build Your Foundation ($0)

- Sign up for ChatGPT (free), Claude (free), Perplexity (free), NotebookLM (free), Google AI Studio (free)
- Use each for one real work task this week — do not just test them with toy examples
- Set up Custom Instructions in ChatGPT with your role, industry, and what you want AI to help with
- Ask Perplexity a research question you would normally Google — compare the quality of answers

## Week 2: Go Deeper

- Create a Claude Project for your most active work area — upload 3-5 reference documents
- Use NotebookLM on a report, contract, or document you need to understand quickly
- Try Copilot in Word or Excel if you have Microsoft 365
- Practice the CRAFT framework on every prompt you write this week

## Weeks 3-4: Build and Create

- Try Replit Agent — describe a simple tool you'd find useful and build it
- Try Midjourney for an image you need for a presentation or marketing asset
- Identify which 2 tools you opened every day during weeks 1-2
- Upgrade to Pro for those 2 tools only — that is your personal AI stack

**Recommended Starting Stack (Week 1, $0)**

ChatGPT (free) — for general writing, analysis, and quick tasks
Claude (free) — for longer documents and detailed reasoning
Perplexity (free) — for research and current information with citations
NotebookLM (free) — for Q&A over your specific documents
Google AI Studio (free) — for huge files and API access

Total cost: $0. These 5 tools alone will transform your productivity.

## AI by Your Career Path

| Your Role | Recommended Tool Combination |
|---|---|
| Finance & Accounting | Copilot Excel + Claude for memos + Perplexity for comps |
| Strategy & Consulting | Perplexity Pro + Claude Projects + Gemini Deep Research |
| Sales & Business Dev | ChatGPT for outreach + Copilot for CRM notes + Gemini research |
| Marketing & Creative | Midjourney + ChatGPT for copy + ElevenLabs + Runway |
| Operations & PM | Copilot Teams + NotebookLM for SOPs + Replit for tools |
| Healthcare / Pharma / Biotech | NotebookLM + Claude 200k context + Perplexity Academic |

**Applied AI Class | Cornell EMBA-Americas 2027**

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*Questions? Reach out. AI is a team sport.*